

2.7 - Inverses and Inverse Trig

Goals: Compute the Derivatives of Inverse functions and Inverse Trig functions.

Before Class Videos: 2.7.1

Note: This worksheet combined some content from inverses in general and trigonometric inverses. The former was covered on Oct 7th by Josh Manzer using his own worksheet and will thus not be covered extensively by Trefor, who will focus on the inverse trig for Friday's Class.

Definition:

For a function $f(x)$, let's define $f^{-1}(x)$ so

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x.$$

Example 1: $y = 2x + 1$

Example 2: $y = x^2$

Example 3: $\frac{d}{dx} \ln x$

Example 4: $\frac{d}{dx}a^x$

Example 5: Let's derive a general formula for the inverse of a function.

Example 6: $\tan^{-1}(\tan x) = x = \tan(\tan^{-1} x)$. Try evaluating these at $x = \pi$.

Example 7: $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Example 8: $\sin^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

Example 9: Plot the graph of $\arcsin(x)$ by comparing to the graph of $\sin(x)$.

Example 10: Compute $\frac{d}{dx} \arcsin(x)$.

Inverse Trigonometric Functions			
Function	Derivative	Domain	Range
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$	$[-1, 1]$	$[0, \pi]$
$\arctan x$	$\frac{1}{1+x^2}$	$\mathbb{R}$	$(-\pi/2, \pi/2)$
$\operatorname{arccot} x$	$-\frac{1}{1+x^2}$	$\mathbb{R}$	$(0, \pi)$
$\operatorname{arcsec} x$	$\frac{1}{ x \sqrt{x^2-1}}$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi/2) \cup (\pi/2, \pi]$
$\operatorname{arccsc} x$	$-\frac{1}{ x \sqrt{x^2-1}}$	$(-\infty, -1] \cup [1, \infty)$	$[-\pi/2, 0) \cup (0, \pi/2]$

The derivatives of $\arcsin x$ and $\arccos x$ hold for $|x| < 1$; those of $\operatorname{arcsec} x$ and $\operatorname{arccsc} x$ hold for $|x| > 1$.

Example 11: $\frac{d}{dx} \sin^{-1}(3x^2 + 1)$

Example 12: Let's do one more derivativion, this time of $y = \sec^{-1}(x)$.

Example 13: $\frac{d}{dx}[\ln(x^2 + 1) + x \tan^{-1}(2x)]$

Example 14: For $\tan^{-1} x + \sin^{-1} y = \frac{\pi}{4}$ find $\left. \frac{dy}{dx} \right|_{(0, \frac{1}{\sqrt{2}})}$